RansKnow: An Open Dataset of Ransomware-Related YouTube
Transcripts with MITRE ATT&CK Annotations

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Abstract

[TODO] Draft, not final. Should state: (1) what RansKnow is (1,128 video records, 1,034 transcripts, 92 cybersecurity YouTube channels, MITRE ATT&CK-aligned regex features); (2) the paper's actual contribution is arguably the label-quality audit methodology as much as the corpus itself – Section 4 found that a naive keyword-matched family label collided with ordinary English words (`play`, `hive`, `cactus`, . . .) badly enough to inflate the single most-detected ransomware family by $\sim$3.5$\times$, and Section 9's counterfactual analysis shows a trained classifier partially inherited that same word-level fragility even after the fix; (3) state plainly, in the abstract itself, not just in a limitations section: baseline results (Section 5) are weak-label agreement scores, not independently verified accuracy – this project does not include human-annotated ground truth (see the scope note in Section 4.3); (4) headline OOD-robustness numbers, which don't depend on that caveat.

1 Introduction

[TODO] The gap: no open, transcript-level ransomware knowledge corpus with MITRE ATT&CK-aligned annotations. Motivate why YouTube security content specifically (volume, diversity of source – vendor threat-intel channels, DFIR conferences, independent researchers – versus e.g. only vendor blog posts). State the paper's contributions as a list: (1) the corpus; (2) the label-quality audit methodology and its concrete findings (Section 4); (3) the experiment framework and weak-label baseline results (Section 5), reported as agreement with the audited extraction pipeline – state directly here, not just in Limitations, that no independent human-annotated ground truth was collected for this project (Section 4.3); (4) out-of-distribution robustness results (Section 8), which are not subject to that caveat; (5) counterfactual and perturbation-robustness interpretability results (Section 9).

2 Related Work

[TODO] Three threads: (a) existing ransomware/threat-intel datasets (structured IOC feeds, MITRE ATT&CK mapping efforts, prior ransomware incident corpora – note what's specific to RansKnow: raw natural-language transcripts rather than structured feeds); (b) weak/distant supervision in security NLP (keyword/regex-based labeling, its known failure modes); (c) conformal prediction and OOD detection in NLP, as background for Sections 7 and 8 once written.

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3 Dataset Construction

RansKnow [Kabuye, 2026] consists of 1,128 video records sourced from 92 cybersecurity YouTube channels, of which 1,034 have an extracted transcript with real content; the remaining 94[1] could not be transcribed by any method in the fetch cascade.

3.1 Channel selection

Channels were curated in two batches. Batch 1 (50 channels, `C01`–`C50`) covers established DFIR vendors, security conferences, and independent researchers. Batch 2 (51 candidate channels, `C51`–`C101`) expanded coverage into threat-intelligence vendors, additional conferences, and OT/ICS-security and government channels; 42 of the 51 candidates yielded usable transcripts, 1 was a confirmed duplicate of an existing Batch 1 channel, 4 had no discoverable dedicated YouTube channel, and 4 resolved to real channels with zero ransomware-keyword-matching content in their upload history.

[TODO] Cite the specific channel registry files (`rubrics/Dataset_Channel_Registry_*.xlsx`) and consider a full channel table as a supplementary table rather than inline.

3.2 Video selection

Within each channel, videos were selected by scanning recent uploads for a fixed keyword list (`ransomware`, `extortion`, `lockbit`, `incident response`, ...; full list in Appendix A) matched against title and description, capped at 20 videos per channel and a minimum duration of 5 minutes.

[TODO] Report the actual keyword list and discuss selection bias explicitly: this is a convenience sample conditioned on channel operators' own titling and description-writing behavior, not a random sample of ransomware discourse.

3.3 Transcript acquisition

Transcripts were acquired via a three-stage cascade: (1) the YouTube Data API's official captions where available; (2) `yt-dlp`'s auto-generated captions; (3) local Whisper transcription as a fallback. **This cascade matters more than a footnote**: only 74 of 1,034 transcripts (7.2%) came from official captions. 959 (92.7%) are ASR output – 554 from `whisper_base` (the smallest, least accurate Whisper checkpoint) and 405 from a local Whisper fallback, 1 from `yt-dlp` captions.

[TODO] This is Finding B from the label-quality audit (Section 4). Decide whether to present it once, here, or duplicate the framing in both sections – probably present the raw fact here and the downstream implication (jargon mistranscription risk) in Section 4.

4 Label-Quality Audit

The Knowledge Agent pipeline (`Scripts/knowledge_agent.py`) extracts structured features from every transcript via regex matching: ransomware family mentions (against a maintained alias list), MITRE ATT&CK tactic mention counts across 10 tactics, offensive tool mentions across 8 tools, and platform signals (Windows/Linux/ESXi). Three concrete problems were found and are described here rather than left as an implicit caveat, because a downstream classifier trained on these labels inherits whatever bias they contain.

[1] **[TODO]** Verify this figure against the live dataset before submission – $1128 - 1034 = 94$, but confirm this is failed transcript extraction and not something else, e.g. videos excluded post-fetch.

4.1 Finding A: generic-word alias collisions

Nine of 41 family aliases collide with common English or business words (`play`, `hive`, `cactus`, `agenda`, `clop`, `inc`, `maze`, `medusa`, `royal`) – identified programmatically via dictionary lookup against `/usr/share/dict/words`, then confirmed by manual transcript inspection (e.g. `hive` matching "registry hive," a routine DFIR term, rather than Hive ransomware). A windowed-context disambiguation fix (requiring a ransomware-context term within ∼15 tokens of the match) was applied and reduces false-positive volume substantially without eliminating it – residual false positives still occur when context vocabulary is itself dense (see `Scripts/audit_family_aliases.py` for the audit methodology and Table 1 for before/after counts).

Table 1: Before/after family mention counts for the 9 flagged aliases, windowed-context disambiguation applied. `Play` alone accounted for 386 of 494 (78%) of all family-tagged videos before the fix.

Family	Before	After	Dropped
Play	386	111	275
Qilin	88	21	67
Hive	23	10	13
Maze	11	5	6
Royal	28	19	9
INC	10	7	3
Cactus	9	6	3
Cl0p	11	11	0
Medusa	7	7	0

4.2 Finding B: ASR-dominated transcript quality

As noted in Section 3.3, 92.7% of transcripts are ASR output rather than official captions.

[TODO] Run the Phase 0.4 spot-check (manual comparison of ASR output against source audio for a fixed jargon term list – `LockBit`, `Mimikatz`, `PsExec`, `Cobalt Strike`, `rclone`, `BloodHound` – across the whisper-transcribed subset) and report a measured word-error-rate-on-jargon figure here. Aggregate detection-rate comparisons across `Transcript_Provider` did not show an obvious gap, which is not the same as no effect – individual mistranscriptions of rare jargon can matter without moving an aggregate rate.

4.3 Scope: no independent ground truth

An earlier design stage of this project planned a held-out, independently annotated evaluation subset (180 videos, stratified by channel type, year, and current family-count) specifically to break the circularity of evaluating a classifier against the same weak-label pipeline it was trained on. That subset was never annotated. External annotator recruitment required institutional ethics approval not available to this project; a single-investigator pass was considered as a fallback and rejected as still too close to the system under evaluation to serve as independent ground truth in a way worth claiming in a paper.

The consequence is stated here plainly, and repeated at the point every affected result is reported, rather than left as a single Limitations bullet a reader could miss: **every quantitative result in Section 5 onward, with the explicit exception of Section 8's out-of-distribution results, measures agreement with the audited Knowledge Agent extraction pipeline (Section 4), not verified accuracy against human judgment.** Section 4 demonstrates concretely that this pipeline's labels contain real, non-trivial error even after the disambiguation fix (Finding A); nothing downstream corrects for that. Readers should treat macro-F1 and calibration numbers in this paper as characterizing how well models reproduce a specific, imperfect, documented extraction process – not as claims about ransomware-domain understanding in an absolute sense. Section 10 returns to this as the paper's central limitation.

5 Task Definitions and Baselines

Five prediction tasks are defined over the corpus: ransomware family (multi-label, 18 classes after excluding 3 singleton-occurrence classes, see Table 2), dominant MITRE tactic (multi-class, 11 classes), platform signal (multi-class, 4 classes), offensive tool mentions (multi-label, 7 classes after excluding `MegaNZ`, which has zero occurrences in the corpus), and a ransomware-relevance binary screen (derived, not a stored column: `Family_Count > 0 OR Tactic_Impact > 0`).

Table 2: Task taxonomy.

Task	Type	Classes	Label source
Family	multi-label	18	alias regex match (audited, §4.1)
Dominant tactic	multi-class (+none)	11	max of 10 tactic-pattern counts
Platform signal	multi-class (+none)	4	max of 3 platform-pattern counts
Tool mentions	multi-label	7	per-tool regex match
Ransomware-relevant	binary	2	derived

5.1 Model ladder and validation protocol

Each task is evaluated over three feature representations (structured Knowledge Agent columns, TF-IDF, embeddings[2]) and five models (the rule-based Knowledge Agent itself as baseline 0, Logistic Regression, Linear SVM, Random Forest, LightGBM), under three cross-validation protocols: stratified random k-fold, `GroupKFold` by `Channel_ID` (to detect channel-style leakage – a model can learn "this is CrowdStrike's narration style" instead of the content), and a temporal split (train pre-2024, test 2024+, since 79% of the corpus is from 2024 onward).

5.2 Results

Per Section 4.3, every number in this section is weak-label agreement – how well a model reproduces the Knowledge Agent's own extraction, not verified accuracy. The rule-based Knowledge Agent baseline scores a tautological macro-F1 of 1.000 on every task (checked against its own labels by construction; included as baseline 0, not as evidence of anything) and is omitted from Table 3 accordingly.

Table 3: Best stratified-CV macro-F1 per task, across every representation (structured / TF-IDF / embedding) and trained model (LogReg, Linear SVM, Random Forest, GBM) tried. Weak-label agreement, not verified accuracy (§4.3).

Task	Best representation	Best model	Macro-F1
Ransomware-relevant	TF-IDF	GBM	0.903
Dominant tactic	structured	GBM	0.813
Platform signal	TF-IDF	GBM	0.606
Tool mentions	TF-IDF	LogReg	0.200
Family	embedding	LogReg	0.129

Two patterns worth naming, both consistent with Section 9's independent checks: (1) tree models (GBM in particular) are the strongest model wherever tested, and structured features remain the strongest representation specifically for `dominant_tactic` – unsurprising, since those features are numerically close to the counts the label itself was derived from; (2) `family` and `tool` are weak under every representation and model combination tried (family never exceeds 0.13 macro-F1; tool is additionally unstable across CV splits, ranging 0.03–0.20). This is not obviously a framework or

[2]

[TODO] Embedding representation is currently blocked in the reference implementation: `sentence_transformers` fails to import due to a torch/transformers version mismatch. Resolve before this table can include embedding-representation numbers.

tuning limitation – Section 9's length and channel-identity confound checks rule out two of the more common alternative explanations for weak tabular/text-classification performance, leaving weak-label ceiling (§4.3) and genuine class sparsity (Section 5's 18-class family / 7-class tool taxonomies, several classes with single-digit support) as the remaining candidate explanations.

[TODO] Full task×representation×model×split table as supplementary material, or as a figure once uncommented below; cite `outputs/experiments/phase1_*.csv` directly.

6 Uncertainty Quantification

Scope note (§4.3 applies here directly): calibration is normally reported against ground truth – does an 80%-confidence prediction turn out correct roughly 80% of the time. Without independent labels, the only calibration target available here is the weak-label distribution itself, which changes what a well-calibrated result would mean: a model perfectly calibrated against its own training signal has learned to be a confident, well-calibrated echo of a regex, not necessarily a well-calibrated model of ransomware content. This section is retained, with that meaning made explicit at every reported number, because reliability diagrams and Expected Calibration Error against the weak labels are still informative about a narrower question worth answering: does a model's confidence track its *agreement with the extraction pipeline*, or is it uniformly overconfident regardless of agreement. The latter would be a red flag on its own.

[TODO] Not implemented yet (Phase 2, weak-label-calibration version). Plan: reliability diagrams + Expected Calibration Error per task/model against weak labels, reported explicitly as such. Random Forest tree-vote entropy as a free uncertainty proxy; MC-Dropout / small deep ensemble for the eventual transformer model. Headline metric: rate of confident (>90%) disagreement with the weak label – the closest analogue available to a "confidently-wrong" rate without ground truth.

7 Conformal Prediction

Scope note: split conformal prediction's distribution-free coverage guarantee is a statement about exchangeability with the calibration set, not about label correctness – it holds formally even when calibrated against weak labels, because the guarantee is "the true *weak* label falls in the predicted set $(1 - \alpha)$ of the time," not "the true *real-world* label does." That distinction is the entire content of this scope note: conformal coverage against the Knowledge Agent's labels is a well-defined, honestly reportable number, but it validates agreement with the extraction pipeline, not correctness. Reported as such.

[TODO] Not implemented yet (Phase 3, weak-label-calibrated version). Plan: split conformal on `dominant_tactic/platform` at 90% target coverage against weak labels; per-label conformal calibration (or conformal risk control) for `family/tool`. Conditional-coverage stress test on post-2025 videos and conference-channel videos (the gold-subset slice from the original design is no longer available, §4.3) – coverage collapse on either remaining slice is itself the result, and feeds directly into the OOD setups below via the nonconformity score.

8 Out-of-Distribution Detection

Unlike Sections 5–7, this section's results are not subject to the weak-label scope note in Section 4.3: OOD detection asks whether a model's own confidence or embedding-space behavior differs between in-distribution and out-of-distribution input, which is answerable without knowing whether either input's *label* is correct.

[TODO] Not implemented yet (Phase 4). Four setups already grounded in real splits: temporal emergence (train pre-2024, test on Qilin/DragonForce/Black Basta, which only appear post-2023 in this corpus), channel-type shift (Vendor+DFIR vs. Conference+Independent), leave-rare-family-out, and an external non-security corpus as a sanity check. Methods to compare: max-softmax-probability, Mahalanobis distance on pooled embeddings, energy score, and the Section 7 nonconformity score used directly as an OOD flag.

9 Interpretability and Perturbation Robustness

Unlike Sections 5–7, this section asks questions about model *behavior* – does it exploit shortcuts, what does it rely on, how much do its predictions move under semantically-irrelevant edits – none of which require independent ground truth to answer, so Section 4.3's scope note does not apply here.

9.1 Confound checks

Length. Transcript word count correlates moderately with `Tactic_Total_Mentions` (Pearson $r = 0.409$) and weakly with `Family_Count` ($r = 0.131$), but not at all with `Tool_Total_Mentions` ($r = 0.043$, not significant) – the confound is not uniform across signal types. Because it is mathematically guaranteed that dividing every tactic's count for one video by that video's own word count cannot change which tactic has the highest count (argmax is scale-invariant per row), the confound cannot affect how the `Dominant_Tactic` label itself was derived; the question that matters is whether it inflates `dominant_tactic`'s strongest result (structured features, Table 3). Re-scoring with length-normalized (mentions per 1,000 words) instead of raw-count features gives 0.753 macro-F1 against the raw-count model's 0.764 – a 1.1-point difference, not a length-exploitation artifact.

Channel identity. 486 of 1,034 transcripts (47%) contain at least one literal mention of their own channel's name. Masking those mentions and re-scoring TF-IDF LogReg under both CV protocols leaves the `channel_grouped`-vs-`stratified_random` gap unchanged for `family` (0.028 raw → 0.031 masked) and for `dominant_tactic` (0.020 raw → 0.029 masked). Whatever a model exploits when a channel is not held out, it is not the channel literally naming itself – the leakage this split protocol exists to catch is stylistic or topical, not addressable by name-masking.

9.2 Counterfactual explanations

Motivated directly by Finding A: does a classifier trained on the *fixed* labels still show the regex's original word-collision fragility? A TF-IDF+LogReg model was trained on today's corrected `Family` labels and evaluated on the 275 videos Section 4.1 confirmed as false positives under the pre-fix regex (videos where "`play`" appears without ransomware context). The trained model assigns essentially zero probability to `Play` on all 275 (maximum $P = 0.013$) – it did not inherit the bare-word-collision bug.

The complementary question – what drives the model's actual positive `Play` predictions – was then asked with a greedy word-*type* removal counterfactual search (remove the vocabulary word whose removal most reduces $P(\text{Play})$, repeat until the prediction flips or a budget is exhausted) on the 8 highest-confidence true positives. `play` appears in 5 of 8 removal sets and is consistently one of the highest-impact single removals when present, but **0 of 8 cases flip on removing "play" alone** – every flip required 4–10 additional removals, most of them generic (function words, not ransomware-specific vocabulary). The model reduced, but did not eliminate, reliance on the single ambiguous trigger word, and the words it combines that reliance with are not obviously a clean, interpretable "ransomware vocabulary" set – worth flagging as its own finding about decision-boundary diffuseness, independent of the word-collision question it was designed to answer.

9.3 Perturbation robustness

Untargeted counterpart to the counterfactual search above: two perturbation types, measuring flip rate (does the top prediction change) and mean absolute probability drift on freshly-fit TF-IDF+LogReg models. *Simulated ASR noise* (compound proper nouns split apart, e.g. `WannaCry`→"wanna cry," `Mimikatz`→"mimi cats," plus 8% random token deletion) was applied to the 74 official-caption videos specifically – the "clean" subset, simulating what those transcripts would look like if they had gone through the ASR cascade that produced the other 92.7% of the corpus (Finding B). *Synonym substitution* (common non-jargon words swapped for synonyms) was applied to a 150-video sample.

dominant_tactic, platform, and relevance are essentially unmoved by either perturbation (0.0–0.7% flip rate, mean drift < 0.01 in every case). family flips 20–26% of the time under both perturbations despite a much smaller mean drift (∼0.002) than any other task – not a new problem, but the same weak-model issue from Table 3 (family's best macro-F1 is 0.129) surfacing differently: closely-bunched, low-confidence probabilities mean a negligible nudge is enough to change which barely-ahead class wins. On the framing of "adversarial" robustness for this dataset: no real adversary submits transcripts to evade this classifier in production, so a full gradient-based adversarial-attack framework would be disproportionate to what this paper needs; the actual risk surface is ASR transcription noise and channel-style variation, which is what this sweep directly measures. Combined with Section 8's temporal-emergence and leave-rare-family-out setups, these results are the paper's answer to how the system handles evolving, novel, and adversarial-adjacent input – not three separate open questions.

10 Limitations and Ethics

[TODO] Expand into full prose; bullets below are the content, not the final form. The first bullet is the paper's central limitation and should probably open this section as prose, not sit inside a list with four other items of noticeably smaller consequence.

- **No independent ground truth.** This is the paper's central limitation, not one bullet among several, and is why it is stated repeatedly rather than once: this project does not include a human-annotated evaluation set independent of the Knowledge Agent pipeline (Section 4.3). External annotator recruitment required institutional ethics approval not available to this project; a single-investigator annotation pass was considered and rejected as still too close to the system under evaluation to count as independent evidence. Every result in Sections 5–7 is consequently agreement with an audited but demonstrably imperfect extraction pipeline (Finding A), not verified accuracy – Section 8's OOD results and Section 9's behavioral diagnostics are the exceptions, since neither depends on label correctness. Recruiting independent annotators, whether through formal ethics approval or an unpaid collaborator arrangement that falls outside whatever definition of human-subjects research applies at the author's institution, is the single highest-value extension of this work – more valuable than any additional modeling in Sections 5–7 would be, since none of that modeling can currently be checked against anything but itself.
- **Sampling bias.** The corpus is conditioned on which channels operators chose to run, which videos they chose to title and describe in keyword-matchable ways, and YouTube's own recommendation and search surfacing – not a random or exhaustive sample of ransomware-relevant discourse.
- **ASR noise.** 92.7% of transcripts are machine-transcribed, mostly by whisper_base. Section 9.3's simulated-ASR-noise sweep bounds how much this moves predictions; Finding B's word-error-rate-on-jargon question (comparing ASR output against source audio directly, rather than a simulation of ASR-style errors) remains open.
- **Weak-label ceiling.** Every quantitative result in Sections 5–7 is bounded by the Knowledge Agent's own precision/recall, which Finding A shows is not perfect even after the disambiguation fix, and which Section 9.2 shows partially propagated into at least one trained model.
- **Dual use.** This is a corpus of ransomware tactics, tools, and case studies extracted from public educational content.

  [TODO] Add a considered dual-use statement – the underlying source material is already public (YouTube videos aimed at defenders), but structuring it may lower the effort to extract offense-relevant patterns; discuss how this compares to existing public MITRE ATT&CK mappings and threat-intel reporting, which already aggregate similar information.

11 Conclusion

[TODO] Write last. Should state the label-quality audit methodology and code (Section 4, Section 9) as a second contribution alongside the transcripts themselves – other weak-label security-NLP datasets built the same way (keyword/regex extraction over scraped text) can reuse the audit methodology even without reusing the corpus. Should also state candidly that the most valuable open follow-up to this specific paper is independent human-annotated verification (Section 10), not further modeling – and should say why explicitly rather than trailing off: every result this paper reports beyond Section 8 would change in *meaning*, not just in number, the day that verification exists.

References

Henry Kabuye. RansKnow: A ransomware knowledge dataset. https://www.kaggle.com/datasets/henrykabuye/ransknow-v1, 2026. Accessed 2026-08-11.

A Keyword List

[TODO] Insert the full `KEYWORDS` list from `Scripts/fetch_transcripts.py` verbatim.

NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction (once finalized – see remaining `\todo` markers) state the corpus, the label-quality audit methodology and findings, weak-label baseline results explicitly framed as pipeline-agreement rather than verified accuracy, and OOD-robustness results as the paper's contributions, matching Sections 3–8.

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2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 10 opens with "No independent ground truth" as the paper's central limitation rather than a minor bullet, and covers sampling bias, ASR transcription noise, the weak-label ceiling, and dual-use considerations.

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4. **Experimental result reproducibility**

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

Answer: [Yes]

Justification: Section 3 specifies the channel/video selection criteria and full keyword list (Appendix A); Section 5 specifies the task taxonomy, feature representations, models, and CV protocols. All extraction, audit, and experiment code is in the public repository (see next item), including the exact hyperparameters used (default scikit-learn/LightGBM settings unless otherwise noted in code).

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  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
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(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: Dataset: `https://www.kaggle.com/datasets/henrykabuye/ransknow-v1` (CC0-1.0). Code: `https://github.com/Henrinnes/RansKnow` (fetch pipeline, label-quality audit scripts, `rk_pipeline` experiment framework, Phase 5 confound/counterfactual/perturbation scripts).

   [TODO] Confirm a repository LICENSE file is present for the code specifically, not just the dataset's CC0 license – add one (MIT recommended, matching the rest of the ecosystem this reuses, e.g. scikit-learn) if missing before submission.

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6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: Section 5 specifies the three CV protocols (stratified random, channel-grouped, temporal) and the five-model/three-representation grid; exact hyperparameters (regularization, tree counts, TF-IDF vocabulary size, embedding model and chunking) are documented in `Scripts/rk_pipeline/` and referenced in Section 5.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

Answer: [No]

Justification: Reported macro-F1 values are single-seed, 5-fold (or single-split, for the temporal protocol) point estimates, not averaged or bounded across repeated random seeds.

[TODO] This is a real, addressable gap, not an inherent limitation – the framework's split protocols already accept a seed parameter; re-running the Phase 1 grid across 3–5 seeds and reporting standard deviation is worth doing before submission, given how cheap most combinations were to run (see compute-resources item below).

Guidelines:

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8. **Experiments compute resources**

Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

Answer: [Yes]

Justification: Experiments ran on a local workstation: 32-core CPU (structured/TF-IDF representations, all sklearn/LightGBM models) plus 3×NVIDIA GTX 1070 Ti, 8GB each (embedding representation, sentence-transformer encoding). Observed wall-clock time: individual task/model/split combinations ranged from seconds to ∼80s for the heaviest (RandomForest on 18-class multilabel TF-IDF); full grids (5 tasks × up to 5 models × 3 splits) completed in 20–45 minutes per representation.

[TODO] State total compute across the whole project, including exploratory runs that didn't make the paper, per the guideline – not tracked precisely during development.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
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Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics https://neurips.cc/public/EthicsGuidelines?

Answer: [Yes]

Justification: The authors have reviewed the NeurIPS Code of Ethics. The corpus is built from already-public YouTube content (Section 3); the current scope includes no human-subjects data collection (Section 4.3); dual-use considerations are discussed in Section 10.

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10. **Broader impacts**

Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

Answer: [Yes]

Justification: Section 10's dual-use bullet discusses both the corpus's positive use (ransomware defense research, DFIR education) and the risk that structuring defender-oriented content could lower the effort to extract offense-relevant patterns.

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Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

Answer: [N/A]

Justification: The dataset contains no operational malware, exploit code, or attack tooling – only transcripts of public educational/defensive security video content. No additional technical safeguards beyond the CC0 open license were implemented.

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    Answer: [Yes]

    Justification: RansKnow is a new dataset, documented via this paper (Sections 3–4) and the Kaggle dataset card.

    [TODO] Croissant machine-readable metadata (core + Responsible AI fields) is required by the Evaluations & Datasets track call for papers and does not exist yet – build and publish alongside the dataset before submission, this is a hard requirement, not optional documentation.

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